



From ZIP to Running: Build Your Alfresco On-Prem Platform

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Agenda

Alfresco Prerequisites

Installation

Moving to External DB



Alfresco Perquisites



Alfresco Architecture

Client
Tier



Web
Browser

Web
Tier



Nginx

Application
Tier



Share



Repository



Content App

Service
Tier



Transform
Service



ActiveMQ



Solr

Data
Tier



PostgreSQL



Content
Store



Solr Index



OpenJDK – Foundation for Alfresco



Apache Tomcat – Application Server for Alfresco



Database – PostgreSQL, MySQL, or MariaDB



Active MQ – Message Broker for Alfresco



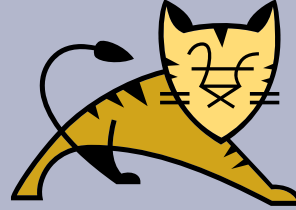
LibreOffice – Document Transformation for Alfresco



ImageMagick – Image Processing for Alfresco



Tying it all together



LibreOffice
The Document Foundation



Student Assigned Servers

User Number	IP Address
Michael	34.229.65.11
Christian	3.238.90.60
Mark	44.210.16.201
Greg	3.239.12.248
Logan	100.49.43.112



Alfresco Installation



<https://short.hyland.com/ACS1CLIVE26>



Script 0

Manual Alfresco ZIP Step	How the Script Automates It
Determine compatible Alfresco component versions	Selects a predefined version profile containing the matching Alfresco, Java, Tomcat, Search, and Transform versions
Record installation settings for later use	Creates a shared <code>alfresco.env</code> file that the remaining installation scripts can load
Define database connection details	Adds the PostgreSQL host, port, database name, username, and generated password
Create service credentials and secrets	Generates random passwords for PostgreSQL, ActiveMQ, the Alfresco administrator, and the Solr shared secret
Configure installation paths and ownership	Detects the current Linux user, group, and home directory automatically
Define component hosts and ports	Stores the Tomcat, Solr, ActiveMQ, Transform, Nginx, Alfresco, and Share connection settings
Prepare the initial administrator password	Generates the administrator password and the hash used during the initial repository bootstrap
Select memory configuration	Documents automatic memory allocation and provides settings that can be overridden before installation



Script 1

Manual Alfresco ZIP Step	How the Script Automates It
Install PostgreSQL	Installs the configured PostgreSQL packages through apt
Create a dedicated Alfresco user	Runs CREATE USER automatically
Assign a password	Reads the password from alfresco.env
Create the Alfresco database	Runs CREATE DATABASE automatically
Use UTF-8 encoding	Includes ENCODING 'UTF8' and verifies it
Grant database permissions	Grants database and schema privileges
Allow password authentication	Updates pg_hba.conf
Restart PostgreSQL	Restarts it with systemctl
Tune the database	Calculates and writes PostgreSQL memory settings



Script 2

Manual Alfresco ZIP Step	How the Script Automates It
Determine the required Java version and server architecture	Detects the system architecture (amd64 or arm64) and uses the configured Java version to select the correct JDK
Check if Java is already installed	Verifies whether the required JDK version already exists before installing
Install the Java JDK	Updates Ubuntu packages and installs the required OpenJDK package automatically
Configure the default Java version	Registers and sets the correct java and javac executables using update-alternatives
Configure JAVA_HOME	Creates a system-wide profile that defines JAVA_HOME and updates the system PATH
Preserve existing configuration	Backs up any existing Java profile before making changes
Verify Java commands	Confirms that both java and javac are available from the command line
Verify the installed version	Ensures the installed Java version matches the version required for Alfresco
Verify the Java installation path	Confirms that the configured JAVA_HOME directory exists



Script 3

Manual Alfresco ZIP Step	How the Script Automates It
Verify Java is installed	Confirms the required Java installation exists and stops if <code>JAVA_HOME</code> cannot be found
Select the correct Tomcat version	Uses the Tomcat version defined in the selected Alfresco version profile
Download the Tomcat package	Downloads Tomcat automatically with a fallback to the Apache archive if needed
Install Tomcat	Extracts Tomcat into the Alfresco installation directory and replaces older versions safely
Configure Tomcat permissions	Assigns ownership to the Alfresco user and secures the installation directories
Configure the Tomcat service	Creates a systemd service with <code>JAVA_HOME</code> , memory settings, keystore configuration, and startup options
Configure automatic startup	Registers and enables the Tomcat service to start automatically on boot
Validate the installation	Verifies the required Tomcat files, service configuration, and installed version
Prevent installation errors	Stops the installation if required files, Java, or service configuration are missing or invalid



Script 4

Manual Alfresco ZIP Step	How the Script Automates It
Verify Java is available	Detects the server architecture, locates JAVA_HOME, and stops if the required Java installation is missing
Select a compatible ActiveMQ version	Uses the ActiveMQ version defined in the selected Alfresco version profile
Download ActiveMQ	Downloads the configured package from the Apache archive and uses the current download server as a fallback
Install the message broker	Extracts ActiveMQ into \${ALFRESCO_HOME}/activemq and backs up a different existing version
Configure file ownership and permissions	Assigns ActiveMQ to the Alfresco account and restricts access to its configuration directory
Configure authentication	Replaces the default broker and web-console credentials with values generated by the configuration script
Configure service access	Updates the web-console bind address and stores the broker host and port settings used by the deployment
Configure ActiveMQ as a service	Creates a systemd service with Java, memory, startup dependency, restart, and security settings
Verify the installation	Confirms the required files, credential permissions, service definition,  enabled status, and installed version

Script 5

Manual Alfresco ZIP Step	How the Script Automates It
Identify compatible component versions	Reads the Alfresco, Search Services, and Transform Core versions from the selected version profile
Locate the required installation packages	Builds the correct Alfresco Nexus URLs for each configured component
Download the Alfresco distribution ZIP	Downloads the Community distribution package containing the repository and Share files
Download Alfresco Search Services	Retrieves the configured Search Services ZIP package
Download the Transform Core package	Retrieves the Transform Core All-in-One JAR
Organize the downloaded resources	Creates a dedicated <code>downloads</code> directory and stores all packages in one location
Avoid unnecessary downloads	Reuses an existing non-empty file instead of downloading it again
Detect failed or corrupt downloads	Checks HTTP results, confirms files are not empty, and validates ZIP and JAR integrity
Record the downloaded versions	Creates a manifest listing the selected versions, filenames, and file sizes 

Script 6

Manual Alfresco ZIP Step	How the Script Automates It
Verify the required software and packages	Confirms Tomcat and the Alfresco distribution are present before making installation changes
Prepare Tomcat for Alfresco configuration	Creates the shared class and library directories and configures Tomcat's shared loader
Extract the Alfresco distribution	Unpacks the downloaded distribution into a temporary working directory
Install repository dependencies	Copies the PostgreSQL JDBC driver into Tomcat's shared library directory
Deploy the Alfresco web applications	Copies the Alfresco and Share WAR files, shared classes, and Tomcat context files into their required locations
Configure storage and security files	Installs the keystore and creates the alf_data, content store, and deleted-content directories
Configure Alfresco services	Generates alfresco-global.properties with database, Search, Transform, ActiveMQ, URL, authentication, and performance settings
Install Alfresco modules	Creates the AMP directories and uses the Module Management Tool to apply repository and Share AMPs



Script 7

Manual Alfresco ZIP Step	How the Script Automates It
Verify Java and Search Services resources	Confirms the correct Java installation and downloaded Search Services ZIP are available
Extract Alfresco Search Services	Unpacks the Search Services distribution into <code>\${ALFRESCO_HOME}/alfresco-search-services</code>
Configure repository connection settings	Adds the Alfresco and Solr host and port values to <code>shared.properties</code>
Configure secure communication	Sets secret-based communication and applies the same shared secret used by the Alfresco repository
Prepare the default Solr cores	Configures Solr to create the <code>alfresco</code> and <code>archive</code> cores during its first startup
Configure the Solr service	Creates a systemd service with Java, Solr Home, memory allocation, and startup commands
Configure startup order and recovery	Starts Solr after Tomcat, requires Tomcat to be available, and restarts Solr after failures
Configure ownership and permissions	Assigns the installation to the Alfresco user and secures the Search Services configuration files



Script 8

Manual Alfresco ZIP Step	How the Script Automates It
Verify Java and the Transform Core package	Detects the system architecture, locates <code>JAVA_HOME</code> , and confirms the downloaded Transform Core JAR exists
Install document-transformation dependencies	Installs ImageMagick, LibreOffice, and ExifTool using the Ubuntu package manager
Prepare LibreOffice for server use	Removes conflicting LibreOffice installations and installs a consistent headless configuration with document, spreadsheet, and presentation support
Install Alfresco PDF Renderer	Downloads the configured PDF Renderer version from Nexus and installs its executable
Install Transform Core AIO	Copies the versioned Transform Core JAR into the Alfresco directory and creates a version-independent symbolic link
Connect the service to ActiveMQ	Configures the broker URL and credentials used by the Transform Service to exchange messages
Configure the Transform service	Creates a systemd service with Java, LibreOffice, PDF Renderer, port, memory, and document-conversion settings
Configure startup and permissions	Runs the service as the Alfresco user, prepares LibreOffice runtime directories, enables startup, and restarts after failures



Script 9

Manual Alfresco ZIP Step	How the Script Automates It
Install the required build tools	Installs the configured Node.js version, npm, and Git when they are not already available
Obtain the ACA source code	Clones the official Alfresco Content App repository into the Alfresco installation directory
Select a compatible ACA release	Checks out either the pinned ACA version or the latest available release tag
Prepare a clean source directory	Fetches updated tags and removes local changes left from earlier builds
Configure the Alfresco backend	Updates the development proxy to use the configured Alfresco repository address
Install project dependencies	Uses the project lock file and npm ci to install consistent Node.js dependencies
Avoid unnecessary dependency installs	Compares the package-lock hash and reuses existing dependencies when they have not changed
Create the production build	Runs the ACA production build and saves the generated files in dist/content-ce
Verify the build output	Confirms the distribution directory, application files, build version, and output size are present



Script 10

Manual Alfresco ZIP Step	How the Script Automates It
Install the Nginx web server	Installs Nginx through the Ubuntu package manager when it is not already available
Prepare a web root for ACA	Creates /var/www/alfresco-content-app as the directory used to serve the application
Deploy the ACA production build	Copies the generated ACA files into the Nginx web root and assigns them to the Nginx service account
Handle a missing ACA build	Creates a temporary information page when ACA has not yet been built
Configure the reverse proxy	Creates an Nginx site that forwards Alfresco Repository and Share requests to Tomcat
Configure ACA browser routing	Serves static files and redirects unknown application routes back to index.html
Support uploads and long-running requests	Removes the Nginx upload-size limit and increases proxy timeouts for large content operations
Enable and validate the Nginx site	Removes the default site, enables the Alfresco configuration, and tests it before applying changes
Configure startup and verify deployment	Enables Nginx at boot, starts or reloads it, and verifies the site, web root, and service configuration 

Script 11

Manual Alfresco ZIP Step	How the Script Automates It
Review the server's available memory	Calculates and displays the memory allocation assigned to the Alfresco services
Start services in the required sequence	Starts PostgreSQL, ActiveMQ, Transform, Tomcat, Solr, and Nginx in a controlled order
Account for an external database	Skips the local PostgreSQL service when Alfresco is configured to use a remote database
Confirm each service is running	Checks the systemd status after issuing each startup command
Verify the database and message broker	Confirms PostgreSQL accepts connections and ActiveMQ is listening on its configured port
Verify Transform and Alfresco readiness	Checks the Transform health endpoint and the Alfresco repository readiness probe
Verify Search Services and Nginx	Pings the Alfresco Solr core with the shared secret and checks the Nginx health endpoint
Handle services that start slowly	Waits for each component, applies service-specific timeouts, and reports initialization delays



Moving to External DB



